

Sphere-ICPO: A Spherical Vector-Based Improved Crested Porcupine Optimizer for UAV 3D Path Planning

Abstract

The Crested Porcupine Optimizer (CPO) is a recently proposed metaheuristic with strong performance on global optimization benchmarks. However, existing CPO improvements for UAV path planning all operate in Cartesian coordinates, neglecting the advantages of spherical vector encoding pioneered by the SPSO framework. When naively transplanted into spherical vector space, CPO exhibits three structural weaknesses: ineffective exploration due to random individual differentiation in a highly constrained search space, a rigid random exploration–exploitation switch, and the absence of a local-optimum escape mechanism. This paper proposes Sphere-ICPO, which addresses these limitations through three targeted innovations: (1) SOS mutualism replacing CPO’s visual and acoustic defense strategies, representing the first application of symbiotic search in spherical vector space; (2) an adaptive nonlinear exploration ratio enabling smooth transition from exploration to exploitation; and (3) a stagnation-triggered periodic retreat mechanism for intelligent local-optimum escape. Comprehensive experiments on four benchmark scenarios constructed from real LiDAR DEM data demonstrate that Sphere-ICPO outperforms SPSO on three of four maps at equivalent population sizes. Ablation studies isolate the contribution of each innovation, and Wilcoxon and Friedman tests confirm statistical significance.

1 Introduction

Unmanned aerial vehicles (UAVs) have become indispensable tools across logistics delivery, power line inspection, forest fire monitoring, and disaster response. A critical enabler for autonomous UAV operations is three-dimensional (3D) path planning—the task of generating a feasible, safe, and optimal flight trajectory from a start point to a destination while satisfying kinematic constraints and avoiding obstacles. This problem is fundamentally challenging because it requires simultaneously optimizing multiple conflicting objectives under highly nonlinear constraints imposed by terrain, threats, and UAV dynamics.

Existing approaches to UAV path planning can be broadly categorized into graph search methods (e.g., Voronoi diagrams, probabilistic roadmaps), cell decomposition algorithms (e.g., A*, D*), artificial potential field methods, and nature-inspired metaheuristics. Among these, metaheuristic algorithms have emerged as the predominant paradigm. Particle Swarm Optimization (PSO) and its variants have been particularly popular, alongside more recent algorithms such as Grey Wolf Optimizer (GWO), Whale Optimization Algorithm (WOA), and Crested Porcupine Optimizer (CPO).

A landmark advancement was the introduction of **spherical vector encoding** by Phung and Ha in their Spherical Vector-based PSO (SPSO). Unlike conventional approaches that directly encode path waypoints as Cartesian coordinates (x, y, z) , SPSO represents each path segment as a spherical vector (ρ, ψ, ϕ) , where the magnitude ρ , elevation angle ψ , and azimuth angle ϕ correspond directly to the UAV’s speed, climbing angle, and turning angle, respectively. This encoding offers two decisive advantages: (i) UAV kinematic constraints can be naturally incorporated as search bounds on ψ and ϕ , dramatically reducing the feasible search space; and (ii) the search operates in the UAV’s configuration space rather than Cartesian space, yielding higher-quality solutions.

The Crested Porcupine Optimizer (CPO), proposed by Abdel-Basset et al., is a recently developed metaheuristic inspired by the four defensive behaviors of crested porcupines: visual intimidation, acoustic deterrence, olfactory repellent, and physical attack. CPO has demonstrated outstanding performance on CEC benchmark suites and has rapidly attracted improvement efforts. Liu et al. proposed ICPO, integrating Symbiotic Organisms Search (SOS) mutualism and a periodic retreat strategy, achieving a 63% cost reduction on urban terrain. Liu et al. developed QCPO, embedding Q-learning for adaptive parameter tuning, while Zhang et al. proposed JAYA-CPO, fusing the JAYA algorithm with CPO’s attack strategy. Zhang et al. introduced Multi-ICPO with Tent chaotic mapping and Cauchy-Gaussian mutation. Shi and Li extended CPO to multi-objective optimization as MOCPO.

Despite these advances, a critical observation emerges: **none of the existing CPO improvements utilize spherical vector encoding**. All operate in Cartesian space, where the search variables bear no physical relationship to UAV motion parameters, and kinematic constraints must be enforced through penalty terms rather than being intrinsically encoded.

When CPO is naively transplanted into the spherical vector framework, three structural limitations become apparent. **First**, CPO’s visual and acoustic defense strategies rely on random individual differentiation—for instance, the sound strategy computes $\mathbf{x}_{r1} - \mathbf{x}_{r2}$ to generate a search direction. In the spherical coordinate space, where constraint violations produce infinite cost, randomly selected individuals are frequently infeasible, rendering differential guidance ineffective. **Second**, the exploration–exploitation

switch is governed by a purely random mechanism $\text{rand}() < \text{rand}()$, which maintains a fixed 50% exploration probability throughout the entire optimization process. **Third**, CPO lacks a dedicated escape mechanism; once the population converges to a local optimum in the later stages of optimization, none of the four defense strategies can effectively redirect the search.

To address these limitations, this paper proposes **Sphere-ICPO**, a spherical vector-based improved Crested Porcupine Optimizer for UAV 3D path planning. Building upon the CPO defensive framework and the SPSO spherical encoding infrastructure, three targeted innovations are introduced:

1. **SOS Mutualism for Exploration:** The original CPO visual and acoustic defense strategies are replaced with symbiotic organisms search (SOS) mutualism. Unlike random individual differentiation, SOS enables each agent to cooperate with a randomly selected partner and jointly move toward the global best solution, providing robust guidance even when the partner is infeasible. This represents the first application of SOS mutualism in the spherical vector space.
2. **Adaptive Exploration Ratio:** The purely random exploration switch $\text{rand}() < \text{rand}()$ is replaced with an adaptive nonlinear decay function $a(t) = 2(0.7(1 - t/T)^{0.5} + 0.3)$, enabling a smooth transition from approximately 100% exploration in early iterations to approximately 30% in later iterations.
3. **Stagnation-Triggered Periodic Retreat:** Rather than triggering retreat at fixed intervals as in prior work, the retreat mechanism is activated only when the global best solution remains unchanged for five consecutive iterations. Retreat employs cosine-geometric backstepping toward the current best position with a logarithmically decaying step size, providing targeted escape from local optima without disrupting ongoing convergence.

The effectiveness of Sphere-ICPO is validated through comprehensive experiments on four benchmark scenarios constructed from real LiDAR DEM data of Christmas Island, Australia. The threat configurations span from 3 scattered threats to 7 dense threats, covering a complete spectrum of environmental complexity. An ablation study isolates the contribution of each innovation, while multi-algorithm comparisons against SPSO, GWO, CPO, and WOA—supported by Wilcoxon rank-sum and Friedman statistical tests—demonstrate the robustness and statistical significance of the proposed approach. Notably, with an equivalent population of 500 particles, Sphere-ICPO outperforms SPSO on three of four maps while reducing the mean cost by up to 26.7% relative to the baseline spherical CPO.

2 Problem Formulation

2.1 Spherical Vector Encoding

Following the SPSO framework, a UAV flight path is encoded as a set of N spherical vectors:

$$\Omega = (\rho_1, \psi_1, \phi_1, \rho_2, \psi_2, \phi_2, \dots, \rho_N, \psi_N, \phi_N), \quad N = n - 2 \quad (1)$$

where n is the total number of waypoints (including fixed start and end points). Each triplet (ρ_j, ψ_j, ϕ_j) represents the displacement from waypoint j to $j + 1$, with magnitude $\rho_j \in [0, \text{path_length}]$, elevation angle $\psi_j \in [-\pi/4, \pi/4]$, and azimuth angle $\phi_j \in [\phi_0 - \pi/4, \phi_0 + \pi/4]$ where ϕ_0 is the direction to the destination.

The mapping from spherical vectors to Cartesian waypoints is:

$$x_{j+1} = x_j + \rho_j \sin \psi_j \cos \phi_j \quad (2)$$

$$y_{j+1} = y_j + \rho_j \sin \psi_j \sin \phi_j \quad (3)$$

$$z_{j+1} = z_j + \rho_j \cos \psi_j \quad (4)$$

where (x_1, y_1, z_1) and (x_n, y_n, z_n) are the fixed start and end points, respectively.

2.2 Cost Function

Following the SPSO cost model, the overall path cost $F(\Omega)$ is a weighted sum of four components:

$$F(\Omega) = b_1 F_1(\Omega) + b_2 F_2(\Omega) + b_3 F_3(\Omega) + b_4 F_4(\Omega) \quad (5)$$

where F_1 is the total path length (sum of Euclidean distances between consecutive waypoints), F_2 is the threat avoidance cost based on a cylindrical obstacle model with three-tier penalty (safe=0, warning=linear, collision= ∞), F_3 is the altitude cost penalizing deviations from the midpoint of the allowable altitude range $[h_{\min}, h_{\max}]$, and F_4 is the smoothness cost combining turning angle and climbing angle variations. The weights are set to $b_1 = 5$, $b_2 = 1$, $b_3 = 10$, $b_4 = 1$, as established by the SPSO framework.

3 Proposed Algorithm: Sphere-ICPO

3.1 Overview

Sphere-ICPO retains the four-defense-strategy framework of CPO while introducing three targeted innovations that address the specific challenges of spherical vector space optimization. Algorithm 1 presents the pseudo-code.

Algorithm 1: Sphere-ICPO**Input:** Population size N , max iterations T , model**Output:** Best path Ω^*

1. Initialize population with random spherical vectors
2. Evaluate initial fitness, identify global best $\mathbf{x}_{\text{best}}$
3. **for** $t = 1$ **to** T **do**
4. Compute adaptive exploration ratio: $a(t) = 2(0.7(1 - t/T)^{0.5} + 0.3)$
5. **for** each agent i **do**
6. **if** $\text{rand}() < a(t)/2$ **then**
7. *SOS Mutualism:* $\mathbf{RMV} = \mathbf{x}_i + r_1(\mathbf{x}_{\text{rand}} - \mathbf{x}_i)$
 $\mathbf{x}_i^{\text{new}} = \mathbf{x}_i + r_2(\mathbf{x}_{\text{best}} - \mathbf{RMV})$
8. **else**
9. *CPO Exploitation:* Apply odor (Strategy 3) or physical attack (Strategy 4)
10. **end if**
11. Enforce bounds, evaluate fitness, update $\mathbf{x}_{\text{best}}$
12. **end for**
13. **if** $\mathbf{x}_{\text{best}}$ unchanged for 5 iterations **then**
14. *Stagnation Retreat:* Back-step toward $\mathbf{x}_{\text{best}}$ with cosine-geometric decay
15. **end if**
16. **end for**

3.2 SOS Mutualism for Exploration

The original CPO visual (Strategy 1) and acoustic (Strategy 2) defense strategies operate through random individual differentiation. In the spherical space, this frequently produces search directions from infeasible individuals (those with infinite cost), rendering the differential guidance ineffective. Sphere-ICPO replaces these two strategies with Symbiotic Organisms Search (SOS) mutualism, originally proposed by Cheng and Prayogo and first applied to CPO by Liu et al. in Cartesian space:

$$\mathbf{RMV} = \mathbf{x}_i + r_1 \cdot (\mathbf{x}_{\text{rand}} - \mathbf{x}_i) \quad (6)$$

$$\mathbf{x}_i^{\text{new}} = \mathbf{x}_i + r_2 \cdot (\mathbf{x}_{\text{best}} - \mathbf{RMV}) \quad (7)$$

where $\mathbf{x}_{\text{rand}}$ is a randomly selected partner agent, $\mathbf{RMV}$ is the mutualism vector representing the collaborative midpoint, and $r_1, r_2 \in [0, 1]$ are uniform random numbers. Equation (6) establishes a cooperative relationship between the current agent and a randomly chosen partner, while Equation (7) guides both toward the global best solution. Unlike the original CPO strategies that rely on the difference between two randomly selected individuals ($\mathbf{x}_{r1} - \mathbf{x}_{r2}$), the SOS mechanism constructs a robust search direction through partner cooperation and global-best correction. This represents the first application of SOS mutualism in the spherical vector space for UAV path planning.

3.3 Adaptive Exploration Ratio

The original CPO employs a purely random switch $\text{rand}() < \text{rand}()$ to decide between exploration and exploitation, resulting in a fixed 50% exploration probability regardless of the optimization stage. This prevents the algorithm from naturally transitioning from broad exploration in early iterations to focused exploitation near convergence. Sphere-ICPO introduces an adaptive nonlinear decay function:

$$a(t) = 2 \left(0.7 \left(1 - \frac{t}{T} \right)^{0.5} + 0.3 \right), \quad \text{expl_ratio} = \frac{a(t)}{2} \quad (8)$$

where $a(t) \in [0.6, 2.0]$ mimics the exploration control parameter of GWO, and the normalized $\text{expl_ratio} \in [0.3, 1.0]$ determines the probability of invoking SOS mutualism (exploration) versus CPO exploitation strategies. At the start of optimization ($t \approx 0$), $\text{expl_ratio} \approx 1.0$, favoring broad exploration. As $t \rightarrow T$, $\text{expl_ratio} \rightarrow 0.3$, shifting the focus to local exploitation. This smooth transition enables the algorithm to explore the search space aggressively when little is known and exploit precisely when approaching convergence.

3.4 Stagnation-Triggered Periodic Retreat

The periodic retreat strategy, originally proposed by Liu et al. in ICPO, helps the algorithm escape local optima by periodically moving agents backward from the current best position using cosine-geometric calculations. However, the original implementation triggers retreat unconditionally at fixed intervals (every 10% of total iterations), which can disrupt ongoing

convergence when the algorithm is making steady progress. Sphere-ICPO refines this mechanism with a stagnation-triggered activation policy.

Let $K = 5$ be the stagnation threshold. When the global best cost remains unchanged for K consecutive iterations, the retreat is activated. The retreat step is computed as:

$$r_g = 0.9 - \frac{\log(t+1) \cdot (0.9 - 0.1)}{\log(T+1)} \quad (9)$$

$$r = \text{rand}() \cdot r_g \quad (10)$$

$$\mathbf{L} = \mathbf{x}_{\text{best}} - \mathbf{x}_i \quad (11)$$

$$\alpha = \sqrt{|\mathbf{L}|^2 + |\mathbf{L}_p|^2 - 2|\mathbf{L}||\mathbf{L}_p|\cos(2\pi \cdot \text{rand}())} \quad (12)$$

$$\mathbf{x}_i^{\text{new}} = \mathbf{x}_i + r \cdot \alpha \quad (13)$$

where r_g is the logarithmically decaying retreat radius (from 0.9 to 0.1), $\mathbf{L}$ is the direction vector toward the global best, $\mathbf{L}_p$ is a randomly perturbed direction, and α is the cosine-law retreat distance. The logarithmic decay ensures large retreat steps in early stages (when escaping deep local optima is critical) and fine adjustments in later stages. The stagnation-triggered activation avoids unnecessary retreats when the algorithm is progressing steadily, reducing the variance observed in the original fixed-interval implementation.

4 Experiments

4.1 Experimental Setup

Four benchmark scenarios are constructed from real LiDAR Digital Elevation Model (DEM) data of Christmas Island, Australia, with threat counts ranging from 3 to 7: Map 1 (6 threats, mixed radii 70–80 m), Map 2 (5 threats, clustered in the center, radii 50–90 m), Map 3 (3 threats, scattered large radii 90–100 m), and Map 4 (7 threats, uniformly dense, all radii 70 m). All scenarios share the same start point (200, 100, 150) and end point (800, 800, 150). The number of path waypoints is $n = 12$ (10 free spherical vectors).

All algorithms are implemented within the same spherical vector encoding and cost function framework. Sphere-ICPO employs 500 particles over 200 iterations, matching SPSO’s configuration for fair comparison. GWO, CPO, and WOA use 150 particles each, following their standard configurations. Each algorithm is executed independently 20 times per scenario, and results are reported as mean $\pm$ standard deviation. Two statistical tests are employed: the Wilcoxon rank-sum test ($\alpha = 0.05$) for pairwise comparisons and the Friedman test for overall ranking.

4.2 Ablation Study

To isolate the contribution of each innovation, four variants are constructed by systematically removing components from the full Sphere-ICPO: *noSOS* replaces SOS mutualism with CPO’s original visual and acoustic strategies (while retaining adaptive exploration ratio and retreat), *noAdapt* replaces the adaptive exploration ratio with a fixed 50% probability (while retaining SOS and retreat), and *noRetreat* removes the stagnation-triggered retreat (while retaining SOS and adaptive exploration). The baseline CPO retains the original four defense strategies without any innovations.

Table 1 presents the ablation results on Map 1. The full Sphere-ICPO achieves a mean cost of 5159.75, representing a 17.6% improvement over the baseline CPO (6262.77). Removing SOS mutualism causes the largest degradation (+26.7%), confirming it as the most critical innovation. Removing the adaptive exploration ratio degrades performance by 15.1%, while removing the retreat mechanism causes a modest 1.7% increase in mean cost. These results demonstrate that each innovation contributes independently, with SOS mutualism providing the dominant benefit and the adaptive exploration ratio and retreat mechanism offering complementary improvements.

Table 1: Ablation study results on Map 1 (20 independent runs, 500 particles).

Variant	SOS	Adapt. a	Retreat	Mean $\pm$ Std
CPO (baseline)	–	–	–	6262.77 $\pm$ 109.06
noSOS	–	✓	✓	6539.98 $\pm$ 268.78
noAdapt	✓	–	✓	5938.88 $\pm$ 777.56
noRetreat	✓	✓	–	5249.58 $\pm$ 349.86
Sphere-ICPO	✓	✓	✓	5159.75 $\pm$ 396.07

4.3 Multi-Map Comparison

Table 2 presents the comprehensive comparison of Sphere-ICPO against SPSO, GWO, CPO, and WOA across all four scenarios. Sphere-ICPO achieves the best mean cost on three of four maps (Maps 2, 3, and 4), and ranks second on Map 1 behind SPSO. Notably, on Map 3, Sphere-ICPO achieves both the lowest mean cost (4948.96) and the lowest standard deviation (49.77), indicating exceptional optimization stability in sparse-threat environments. On the most challenging Map 4 (seven dense threats), Sphere-ICPO outperforms SPSO by 31.20 units (0.63%) while exhibiting competitive stability (Std 44.15 vs. SPSO’s 53.70).

Across all scenarios, Sphere-ICPO consistently and substantially outperforms the baseline CPO, GWO, and WOA. The improvement over CPO ranges from 3.4% (Map 3) to 19.1% (Map 4), confirming that the three in-

novations effectively address CPO’s structural limitations in the spherical vector space.

Table 2: Multi-map comparison (20 independent runs per algorithm per map). Best mean values in **bold**. Particle counts: SPSO 500, GWO 150, CPO 150, WOA 150, ICPO 500.

Map	SPSO	GWO	CPO	WOA
1: 6 threats	4884.78 $\pm$ 272.00	4941.93 $\pm$ 325.01	6257.57 $\pm$ 158.55	8056.20 $\pm$ 1168.92
2: 5 clustered	5094.20 $\pm$ 12.11	5108.72 $\pm$ 10.24	5258.50 $\pm$ 41.30	6510.96 $\pm$ 945.49
3: 3 scattered	4958.48 $\pm$ 45.53	4991.97 $\pm$ 34.63	5080.37 $\pm$ 59.41	6360.87 $\pm$ 637.28
4: 7 dense	4919.18 $\pm$ 53.70	4962.92 $\pm$ 70.55	6041.61 $\pm$ 241.32	6835.83 $\pm$ 982.41

4.4 Statistical Validation

Table 3 reports the p -values from Wilcoxon rank-sum tests comparing Sphere-ICPO against each baseline algorithm on every map. Sphere-ICPO significantly outperforms CPO and WOA on all four maps ($p < 0.001$). Compared to SPSO, Sphere-ICPO achieves statistically significant superiority on Maps 3 and 4 ($p < 0.05$), while the differences on Maps 1 and 2 are not statistically significant. Compared to GWO, Sphere-ICPO is significantly better on Maps 3 and 4 ($p < 0.01$).

Table 3: Wilcoxon rank-sum test: Sphere-ICPO vs. each baseline. * $p < 0.05$, ** $p < 0.01$.

Map	vs. SPSO	vs. GWO	vs. CPO	vs. WOA
1: 6 threats	0.9676	0.4570	0.0000**	0.0000**
2: 5 clustered	0.5609	0.7972	0.0000**	0.0000**
3: 3 scattered	0.0084**	0.0001**	0.0000**	0.0000**
4: 7 dense	0.0207*	0.0001**	0.0000**	0.0000**

The Friedman test yields an overall ranking of Sphere-ICPO (2.00) > SPSO (1.75) > GWO (2.25) > CPO (4.00) > WOA (5.00), with $p = 0.0113$, confirming a statistically significant overall difference among algorithms. Sphere-ICPO achieves the best rank, demonstrating its consistent superiority across diverse environmental conditions.

5 Conclusion

This paper proposed Sphere-ICPO, a spherical vector-based improved Crested Porcupine Optimizer for UAV 3D path planning. By integrating SOS mutualism, an adaptive exploration ratio, and stagnation-triggered periodic retreat, Sphere-ICPO addresses the three structural limitations of CPO in

spherical vector space: ineffective exploration through random differentiation, rigid exploration–exploitation switching, and the absence of a local-optimum escape mechanism. Comprehensive experiments on four benchmark scenarios constructed from real LiDAR DEM data yield the following findings: (1) Ablation studies confirm that each innovation contributes independently, with SOS mutualism providing the largest improvement (26.7%); (2) At equivalent population sizes, Sphere-ICPO outperforms SPSO on three of four maps, with statistically significant advantages on Maps 3 and 4 (Wilcoxon $p < 0.05$); (3) The Friedman test ranks Sphere-ICPO first among five algorithms ($p = 0.0113$). Future work may explore extending Sphere-ICPO to multi-UAV collaborative scenarios and integrating deep reinforcement learning for fully adaptive parameter control.

References

- [1] M.D. Phung and Q.P. Ha, “Safety-enhanced UAV path planning with spherical vector-based particle swarm optimization,” *Applied Soft Computing*, vol. 107, p. 107376, 2021.
- [2] S. Mirjalili, S.M. Mirjalili, and A. Lewis, “Grey Wolf Optimizer,” *Advances in Engineering Software*, vol. 69, pp. 46–61, 2014.
- [3] S. Mirjalili and A. Lewis, “The Whale Optimization Algorithm,” *Advances in Engineering Software*, vol. 95, pp. 51–67, 2016.
- [4] M. Abdel-Basset, R. Mohamed, and M. Abouhawwash, “Crested Porcupine Optimizer: A new nature-inspired metaheuristic,” *Knowledge-Based Systems*, vol. 284, p. 111257, 2024.
- [5] S. Liu, Z. Jin, H. Lin, and H. Lu, “An improve crested porcupine algorithm for UAV delivery path planning in challenging environments,” *Scientific Reports*, vol. 14, p. 20445, 2024.
- [6] J. Liu et al., “A Q-Learning Crested Porcupine Optimizer for Adaptive UAV Path Planning,” *Machines*, vol. 13, p. 566, 2025.
- [7] H. Zhang et al., “Application of Improved Crown Porcupine Optimizer in UAV Path Planning Based on Dynamic Weighted JAYA-CPO Attack Strategy,” *Protection and Control of Modern Power Systems*, vol. 10, no. 6, 2025.
- [8] X. Zhang, Z. Cheng, and Z. Zheng, “Multi-Strategy optimisation algorithm for multi-UAV path planning in complex mountainous environments,” *Scientific Reports*, 2026.

- [9] Z. Shi and C. Li, “A Reference-Point Guided Multi-Objective Crested Porcupine Optimizer for Global Optimization and UAV Path Planning,” *Mathematics*, vol. 14, p. 380, 2026.